Assignment Overview: Data Migration to New Architecture

Estimated time: 4 minutes

Learning objectives

- Apply data migration techniques to transfer data to the new architecture
- Design appropriate schemas and migration strategies that align with the EDA standards during the transition

Introduction

In phase 1, you thoroughly assessed the existing data architectures of **FashionMart** and **TrendyThreads**. This involved examining their data storage systems, integration mechanisms, data quality processes, security protocols, compliance standards, and governance frameworks. You documented the current architectures, identified silos, inefficiencies, and bottlenecks, and proposed opportunities for improvement based on best practices.

In phase 2, you applied the insights gained from the assessment to design the future data architecture for the merged organization, **FutureMart**. You created architectural diagrams for the new databases, developed ER diagrams and table designs, and built initial schemas. Your design addressed previously identified gaps and inefficiencies, setting the foundation for a scalable and efficient data infrastructure.

In this module, you will work on the tasks of phase 3.

Phase 3: Develop and implement data migration strategy

In this phase, you will focus on building and executing a comprehensive data migration strategy for **FutureMart**. You'll plan data migration between **TrendyThreads'** digital-first systems and **FashionMart's** traditional infrastructure. This includes transferring data across relational RDBMS and NoSQL databases, ensuring data integrity, compatibility, and consistency. The goal is to seamlessly transition to the newly designed architecture without disrupting ongoing business operations.

Tasks:

Exercise 1: Migrate product catalog data from RDBMS to NoSQL

In this exercise, you will migrate product catalog data from a MySQL relational database to a MongoDB NoSQL database.

- **Initialize the source RDBMS database:** Set up the source MySQL database environment.
- **Perform data migration:** Use the appropriate migration tool or script to migrate the product catalog data from MySQL to MongoDB.
- **Validate the migration:** Verify that the migration was successful.

Exercise 2: Migrate product catalog data from NoSQL to RDBMS

In this exercise, you will migrate structured product catalog data from MongoDB to MySQL.

- **Initialize the source NoSQL database:** Set up the MongoDB environment with sample data
- **Perform data migration:** Migrate the data from MongoDB to MySQL
- **Validate the migration:** Ensure successful data transfer

Deliverables

By the end of **Module 3**, you will have completed the following deliverables:

- Screenshot **RDBMS_to_NoSQL.jpg** showing successful data migration to MongoDB
- Screenshot **NoSQL_to_RDBMS.jpg** showing successful data migration to MySQL